Related Rates

Section 1: Geometric Expansion (Circles, Spheres, Cubes)

1 The radius of a circle expands at a constant rate of 3 cm/s. How fast is the area increasing when the radius is 10 cm?

a. 60π cm^2/s

b. 30π cm^2/s

c. 100π cm^2/s

d. 20π cm^2/s

2 Air is pumped into a spherical balloon so that its volume increases at 100 cm^3/s. How fast is the radius increasing when $r = 5$ cm?

a. $\frac{1}{\pi}$ cm/s

b. $\frac{2}{\pi}$ cm/s

c. $\frac{4}{\pi}$ cm/s

d. 5π cm/s

3 The edge length s of a cube increases at 2 cm/s. At what rate is the surface area increasing when $s = 4$ cm?

a. 96 cm^2/s

b. 48 cm^2/s

c. 32 cm^2/s

d. 64 cm^2/s

Section 2: Sliding Ladders and Pythagorean Triangles

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4 A 13-ft ladder leans against a vertical wall. If the bottom slides away from the wall at 2 ft/s, how fast is the top sliding down when the bottom is 5 ft from the wall?

a. 5/6 ft/s

b. 12/5 ft/s

c. 2 ft/s

d. 1/2 ft/s

5 Two cars start from the same point. Car A travels north at 40 mph and Car B travels east at 30 mph. How fast is the distance between them increasing after 2 hours?

a. 50 mph

b. 70 mph

c. 45 mph

d. 60 mph

6 In the 13-ft ladder problem ($x = 5, y = 12, dx/dt = 2$), at what rate is the area of the triangle formed by the ladder changing?

a. 119/12 ft^2/s

b. 25/2 ft^2/s

c. 0 ft^2/s

d. 12 ft^2/s

Section 3: Conical and Cylindrical Tanks

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7 Water flows into an inverted conical tank (radius 4 m, height 12 m) at 2 m^3/min. How fast is the water level rising when the depth is 6 m?

a. $\frac{1}{2\pi}$ m/min

b. $\frac{1}{\pi}$ m/min

c. $\frac{2}{\pi}$ m/min

d. $\frac{1}{4\pi}$ m/min

8 A cylindrical tank of radius 3 m is being filled at 18π m^3/min. How fast is the water height increasing?

a. 2 m/min

b. 3 m/min

c. 6 m/min

d. 1 m/min

9 A trough with triangular cross-section (base 2 m, depth 2 m, length 6 m) leaks water at 0.5 m^3/min. What is dh/dt when $h = 1$ m?

a. $-1/12$ m/min

b. $-1/6$ m/min

c. $-1/8$ m/min

d. $-1/4$ m/min

Section 4: Shadows and Angles of Elevation

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10 A 6-ft tall person walks away from a 15-ft streetlight at 4 ft/s. At what rate is the length of the person's shadow increasing?

a. 8/3 ft/s

b. 2 ft/s

c. 4 ft/s

d. 10/3 ft/s

11 In the same streetlight scenario, at what rate is the tip of the shadow moving?

a. 20/3 ft/s

b. 8/3 ft/s

c. 4 ft/s

d. 12 ft/s

12 An observer 100 m from a rocket launch site watches a rocket rise vertically at 50 m/s. Find $\frac{d\theta}{dt}$ when the rocket is 100 m high.

a. 0.25 rad/s

b. 0.50 rad/s

c. 1.00 rad/s

d. 0.10 rad/s

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